10

1.

$$\begin{cases} L = \frac{1}{2}mv^2 + ev\cdot A - e\phi \\ p = mv + eA \\ H = pv - L \end{cases} =$$

Gauge :

$$\begin{cases} A \to A + \nabla\chi \\ \phi \to \phi - \partial_t\chi \\ \psi \to \psi e^{i\frac{e}{\hbar}\chi} \end{cases}$$

A - B effect

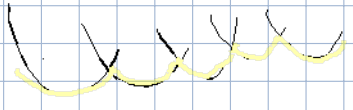
$$H = \frac{(p - eA)^2}{2m}$$

$$\psi_1 = e^{i\theta}$$

Generalized :

$$H = \frac{(p - eA)^2}{2m}$$

$$\psi = \exp\left(i\frac{S}{\hbar} + i\frac{e}{\hbar}\int_0^r A\cdot ds\right)$$



2. Application

(1) Superconductivity.

① Little-Park

$$H = \frac{|(-i\hbar\nabla - e^*A)\psi|^2}{2m^*}$$

$$\text{실험} \to \lambda \sim \frac{\hbar}{e^*} \Rightarrow e^* = 2e$$

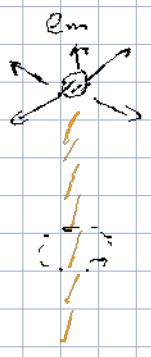
② Flux Quantization

$$J = \frac{e^* n^*}{m^*}(\hbar\nabla\phi - e^*A)$$

$$\oint A\,dl = \frac{h}{2e}$$

(2) Dirac Monopole.

$$\begin{cases} \nabla \cdot \vec{E} = \dfrac{e\delta(\vec{r})}{\varepsilon_0} \\ \nabla \cdot \vec{B} = e_m \delta(\vec{r}) \end{cases} \rightarrow \begin{cases} \vec{E}(\vec{r}) = \dfrac{e}{4\pi\varepsilon_0 r^2}\hat{r} \\ \vec{B}(\vec{r}) = \dfrac{e_m}{4\pi r^2}\hat{r} \end{cases}$$



what kind of Ω : $\nabla \times \vec{\Omega} = \vec{B}$?

If $\vec{\Omega} = \Omega_\varphi \hat{\varphi}$ $\quad \nabla = \cdots$

$$\frac{\partial_\theta (\Omega_\varphi r \sin\theta)}{r^2 \sin\theta} = \frac{e_m}{4\pi r^2} \rightarrow \Omega_\varphi = \frac{e_m}{4\pi r} \frac{C - \cos\theta}{\sin\theta}$$

Let $C = 1 \rightarrow \Omega_\varphi = \dfrac{e_m}{4\pi r} \tan\left(\dfrac{\theta}{2}\right)$

$\theta = \pi$ is a singularity. Dirac String.

Arbitrary C → Any pointing of DS

A loop around DS

$$\frac{e}{\hbar} \oint \Omega \cdot d\vec{l} = \frac{e}{\hbar} \frac{e_m}{4\pi r} \frac{1-\cos\theta}{\sin\theta} 2\pi r$$

$$= \frac{e e_m}{\hbar}$$

Wave Function Continuous → $\dfrac{e e_m}{\hbar} = 2n\pi$. Charge Quantization.

(3) Gravitational AB effect.

$$\frac{1}{\hbar}\int_{ABD} V dt - \frac{1}{\hbar}\int_{ACD} V dt =$$

$$= \frac{m^2 g^2 l^2 r \lambda}{2\pi\hbar^2} \sin\delta$$

3.

$$\begin{cases} \text{AB phase} & \dfrac{e}{\hbar}\oint \vec{A}\cdot d\vec{l} = 2\pi \cdot \dfrac{\Phi}{\Phi_0} \\ \text{winding num} & \psi \sim e^{i n\theta} \cdot \oint \nabla\alpha \cdot d\vec{l} = 2\pi n \end{cases}$$

4. Adiabatic Theorem

$H(\lambda(t))$

$$|\psi(t=0)\rangle = \sum_n c_n |\psi_n(\lambda(0))\rangle$$

$$|\psi(t)\rangle = \sum_n c_n \exp\left\{-\frac{i}{\hbar}\int_0^t E_n(\lambda(\tau))\,d\tau\right\} |\psi_n(\lambda(t))\rangle$$

Dynamic Phase

$$= \sum_n a_n e^{-i\phi_n} |n\rangle$$

↳ Schrodinger Eq.

$$i\hbar\,|\dot\psi(t)\rangle = i\hbar \sum_n \left[\dot c_n e^{-i\phi_n}|n\rangle + c_n(-i\dot\phi_n) e^{-i\phi_n}|n\rangle + c_n e^{-i\phi_n}\partial_t|n\rangle\right]$$

$$= i\hbar \sum_n \left[\dot c_n e^{-i\phi_n}|n\rangle + c_n\left(-i\frac{E_n}{\hbar}\right) e^{-i\phi_n}|n\rangle + c_n e^{-i\phi_n}\partial_t|n\rangle\right]$$

$$= \sum_n c_n e^{-i\phi_n} E_n |n\rangle$$

$$\rightarrow \sum_n \left[\dot c_n e^{-i\phi_n} + c_n e^{-i\phi_n}\partial_t\right)|n\rangle$$

$\langle m|$

$$\dot c_m e^{-i\phi_m} + \sum_n c_n e^{-i\phi_n}\langle m|\partial_t|n\rangle = 0$$

For $n \neq m$:

$$\partial_t(\langle n|n\rangle) =$$

$$\dot c_n + c_n\langle n|\partial_t|n\rangle = 0$$

Berry Phase : $c_n = e^{i\theta_n(t)}$

$$\left\{i\dot\theta_n(t) + \langle n|\partial_t|n\rangle\right\} e^{i\theta_n(t)} = 0$$

$\partial_t \to d\tau$

$$\theta_n(t) = \int_0^T \langle n|i\partial_\tau|n\rangle\,d\tau$$

$\downarrow$

$\partial_\lambda \to d\lambda$

$$\theta_n(\lambda) = \int_{\lambda_0}^{\lambda} \langle n|i\partial_\lambda|n\rangle\,d\lambda$$

gauge-dep.

For a closed loop C :

$$\theta_n(C) = \oint_C \langle n|i\partial_\lambda|n\rangle\,d\lambda$$

gauge-indep.

3. Berry Connection & Curvature.

$$\mathcal{A}_\mu^{(n)} = i\langle n(\mathbf{k})|\partial_k|n(\mathbf{k})\rangle, \quad \partial_\mu = \frac{\partial}{\partial k_\mu}$$

$$\gamma_n(C) = \oint_C \vec{\mathcal{A}}_n \cdot d\vec{k} = \iint (\partial_\mu \mathcal{A}_{n\nu} - \partial_\nu \mathcal{A}_{n\mu})\, dk_\mu dk_\nu \quad (\mu<\nu)$$

$$= \iint \Omega_{n,\mu\nu}\, dk_\mu dk_\nu \quad \text{Berry Curvature.}$$

$$\Omega_{n,\mu\nu} = \partial_\mu \mathcal{A}_{n\nu} - \partial_\nu \mathcal{A}_{n\mu}$$

$$= i(\partial_\mu \langle n|\partial_\nu|n\rangle - \partial_\nu \langle n|\partial_\mu|n\rangle)$$

$$= i(\partial_\mu \langle n|\partial_\nu n\rangle - \partial_\nu \langle n|\partial_\mu n\rangle)$$

$$= i(\langle \partial_\mu n|\partial_\nu n\rangle - \langle n\partial_\mu\partial_\nu|n\rangle - \langle \partial_\nu n|\partial_\mu n\rangle + \langle n\partial_\nu\partial_\mu|n\rangle)$$

$$=$$

Example : 1/2 spin

$$\psi = \begin{pmatrix} \frac{\partial}{\partial \sin\theta} & \\ & e^{-i\phi} \frac{\partial}{\partial \cos\theta} \end{pmatrix}$$

$$\langle u | \theta \rangle \langle \theta | s | u \rangle = \theta_{uv}$$

$$= \left| \left(\frac{\epsilon^{(1)}}{\epsilon^{(2)}} \right) \cos\frac{\theta}{2} \langle \hat{e}_\theta \rangle \sin\frac{\theta}{2} + \left(\frac{\epsilon^{(1)}}{\epsilon^{(2)}} \right) \sin\frac{\theta}{2} \langle \hat{e}_\theta \rangle \cos\frac{\theta}{2} e^{i\phi} \right| + =$$

$$= 0$$

$$\langle u | \partial_\phi | u \rangle = \partial_\phi$$

$$= \frac{i}{\epsilon} \left| \left(\frac{\epsilon^{(1)}}{\epsilon^{(2)}} \right) \sin^2\frac{\theta}{2} - \left(\frac{\epsilon^{(2)}}{\epsilon^{(1)}} \right) \cos^2\frac{\theta}{2} \right| e^{-i\phi} \partial_\phi \langle s | -$$

Example 2: spin-1/2

① $|u(\theta,\phi)\rangle = \langle \theta,\phi | u \rangle$

$$\langle m' | R_{l'}^{\dagger}(\theta) R_{\phi}^{\dagger}(\phi) \partial_\theta R_\phi(\phi) R_{l'}(\theta) | m \rangle = d\theta$$

$$= 0$$

$$\langle m' | R_{l'}^{\dagger}(\theta) R_\phi(\phi) \partial_\phi R_\phi^{\dagger}(\phi) R_{l'}(\theta) | m \rangle = d\phi$$

$$= m \cos\theta$$

② $\Omega_{n,m,3} = \frac{1}{2}$

$\Omega_{n,m,3} = \Omega_{m,n}$;



$\Omega_{m,n} = \dots = 0$

$\Omega_{m,11} = \dots = 0$

$\Rightarrow \vec{\Omega}_m = -\frac{m}{R^2}\hat{R}$.

$\sigma_m = \iint \vec{\Omega}_m \cdot d\vec{S} = -m \cdot \Omega$ — solid angle.

Comparison: Spin on sphere / Dirac's monopole.

1st Chern Number. $C = \frac{1}{4\pi} \iint \vec{\Omega} \cdot d\vec{S}$

6. Physical Meaning.

(1) 2D QHE

$$\begin{cases} \dot{\vec{r}} = \frac{\partial \varepsilon_{\vec{k}}}{\hbar \partial \vec{k}} - \dot{\vec{k}} \times \Omega_{\vec{k}} \\ \hbar \dot{\vec{k}} = -e\vec{E} - e\dot{\vec{r}} \times \vec{B} \end{cases}$$ Lorentz Force in k-space

$$\vec{J} = \int f_{\vec{k}}(-e)\dot{\vec{r}}\, d\vec{k}$$

$$= \square - e \int_{\vec{k}} \dot{\vec{k}} \times \Omega_{\vec{k}} f_{\vec{k}} = \square + \frac{e^2}{\hbar} \int_{\vec{k}} \vec{E} \times \Omega_{\vec{k}} f_{\vec{k}}$$

$$\sigma_{xy} = \frac{e^2}{\hbar} \iint_{\vec{k}} \Omega_{z\vec{k}} f_{\vec{k}} = \frac{e^2}{\hbar} C$$ Anomalous QHE

(2) TKNN

$\vec{B}$ exist → unit cell ↗ → BZ ↘ → more band in BZ

7. Remark